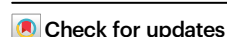


Patterned electrical brain stimulation by a wireless network of implantable microdevices

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Transmitting meaningful information into brain circuits by electronic means is a challenge facing brain-computer interfaces. A key goal is to find an approach to inject spatially structured local current stimuli across swaths of sensory areas of the cortex. Here, we introduce a wireless approach to multipoint patterned electrical microstimulation by a spatially distributed epicortically implanted network of silicon microchips to target specific areas of the cortex. Each sub-millimeter-sized microchip harvests energy from an external radio-frequency source and converts this into biphasic current injected focally into tissue by a pair of integrated microwires. The amplitude, period, and repetition rate of injected current from each chip are controlled across the implant network by implementing a pre-scheduled, collision-free bitmap wireless communication protocol featuring sub-millisecond latency. As a proof-of-concept technology demonstration, a network of 30 wireless stimulators was chronically implanted into motor and sensory areas of the cortex in a freely moving rat for three months. We explored the effects of patterned intracortical electrical stimulation on trained animal behavior at average RF powers well below regulatory safety limits.

Electrical stimulation of the nervous system has a venerable history. Many important discoveries have led to valuable clinical therapies such as deep brain stimulation to manage Parkinsonian conditions^{1–3}, or the use of surface stimulation to identify eloquent areas prior to neurosurgery^{4–6}. In the contemporary field of brain-computer interfaces (BCI), a key aspiration is to advance electronic means to write meaningful information into the cortex. Major advances have been made in recording cortical signals, which once successfully decoded, have enabled recent demonstrations of paralyzed people communicating via keyboard commands^{7–9} and translating verbal motor intentions to machine-generated speech^{10–12}. For a future closed-loop BCI system, however, the ability to provide meaningful and naturalistic electrophysiological feedback directly into the brain in real time remains an elusive goal, in part because of the lack of a clinically viable neurotechnology.

In the case of focused electrical stimulation, there is considerable evidence that current pulses injected locally, e.g., into the hand area of the somatosensory cortex, can elicit tactile sensation in non-human^{13–16} and human primates^{17–19}. Injecting retinotopically matched current stimulus into the primary visual cortex of a non-human primate has also been reported to elicit distinct visual corollaries^{20–22}. In these works, silicon microelectrode arrays (MEAs), such as the Utah array^{23–25}, have been adopted to deliver simple patterns of current stimulation in assessing the efficacy of microstimulation. However, such monolithic tethered implants have drawbacks in fixed electrode configuration, a finite number of channels, and hence limited cortical coverage. For example, in early clinical trials^{26–28}, an MEA evoked sensation only in part of one or two fingers^{26,28}. Scaling up to cover additional cortical areas and implanting multiple MEAs may be impractical for long-term chronic use. Likewise, percutaneous

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connections are undesirable, highlighting the importance of a wireless implant. We note that delivering focal currents into the cortex has recently been achieved in microprobe designs using magnetic induction as the mechanism^{29,30}.

Recently, a wireless multichannel cortical stimulator motivated by prospects for visual prostheses was introduced, which enables the delivery of patterns of current through a monolithic microelectrode array^{31,32}. In this particular embodiment, a coil, an application-specific integrated circuit (ASIC), and a microelectrode array were heterogeneously assembled on a ceramic substrate, resulting in a relatively large form factor for a chronic implant. Thus, there is impetus to engineer a scalable, versatile, and wireless miniaturized neurostimulation system, which is capable of delivering possibly complex spatiotemporal patterns at a large number of cortical points and across multiple cortical areas.

As an alternative to monolithic multielectrode arrays, autonomous free-standing wireless microchips have been suggested as an avenue toward BCI-related stimulatory interventions^{33–36}. Khalifa et al. reported a miniaturized version of a wireless neural implant based on monolithic silicon integrated circuits aimed at stimulating the cortex or peripheral nerves³⁴. However, the design was limited to a single device/channel due to its radiofrequency (RF) communication configuration. In a related scheme, an ultrasound-powered neural microstimulator was introduced for peripheral nerve stimulation³⁵. Ultrasound powering allows for deeper penetration into soft tissue than RF; however, the size of the stimulator device remained relatively large because of constraints imposed by the heterogeneous material assembly. In this case, too, only a single device was demonstrated in an anesthetized rodent model. Elsewhere, an endovascular neural stimulator housing a magnetoelectric film has been reported, constrained to a single device so far³⁶. A representative summary of contemporary efforts to develop wireless neural microstimulators is given in Supplementary Table 1.

In the context of neural recording, our group previously introduced the idea of spatially distributed autonomous silicon microchips –“neurograins”–as a readily scalable wireless neural recording technology, which might in principle be complemented with the capability for intracortical stimulation^{37–40}. In this paper, we demonstrate the proof-of-concept technology for a scalable multipoint wireless electrical stimulation platform by introducing an advanced class of implantable, remotely programmable system-on-chip (SoC) silicon microchips. Power to the sub-millimeter-sized microdevices (300 μm to 500 μm in linear dimension) is harvested from an external RF source near 1 GHz and converted into pulsed biphasic currents of up to 120 μA peak-to-peak, deliverable to a specific target cortical area in which each particular chip is implanted.

A key innovation in the paper is the development of a method to program the stimulation parameters (current amplitude, period per phase, and repetition rate) across a large network of spatially distributed chips in real time. We conceived a custom collision-free low-duty-cycle communication protocol, a wireless register-mapping method denoted as a ‘daisy-chain’ protocol. Below, we demonstrate the system implemented as a chronic implant in a freely moving rodent delivering stimulation simultaneously through up to thirty microstimulators. In the communication scheme, the external RF transmitter issues pre-scheduled commands across the chip population via a high-speed downlink at 1 Mbps, addressing and acknowledging the current release events from each device at rates, which are instantaneous on a neurophysiological timescale. The approach allows us to control a network of a large number of microdevices and transmit wireless power to generate space-time patterned stimulation only during the delivery of an actual pulsed current stimulus. This significantly reduces the total average RF power exposure, ensuring it remains well below regulatory safety limits, a major advantage of the proposed system.

We assessed the performance of the full wireless system in two in vivo phases. Following benchtop validation by injecting currents

from a population of microchips into saline, we applied patterns of intracortical microstimulation in anesthetized rats and recorded the evoked neural responses over a range of stimulus conditions. We then implanted 30 microchips across the cortex of a rat for chronic experiments and applied spatially targeted stimulation while selectively sampling the vast stimulus parameter space in a freely moving animal over three months. Distinct neuromodulation effects were measured as the animal engaged in a trained lever-pressing task. Lastly, although the size of the rat brain limits the number of implants, extrapolation of the experimental data suggests that a wireless system composed of up to 1000 microstimulators can be accessed in less than 3 milliseconds to deliver complex dynamical patterns of cortical excitation.

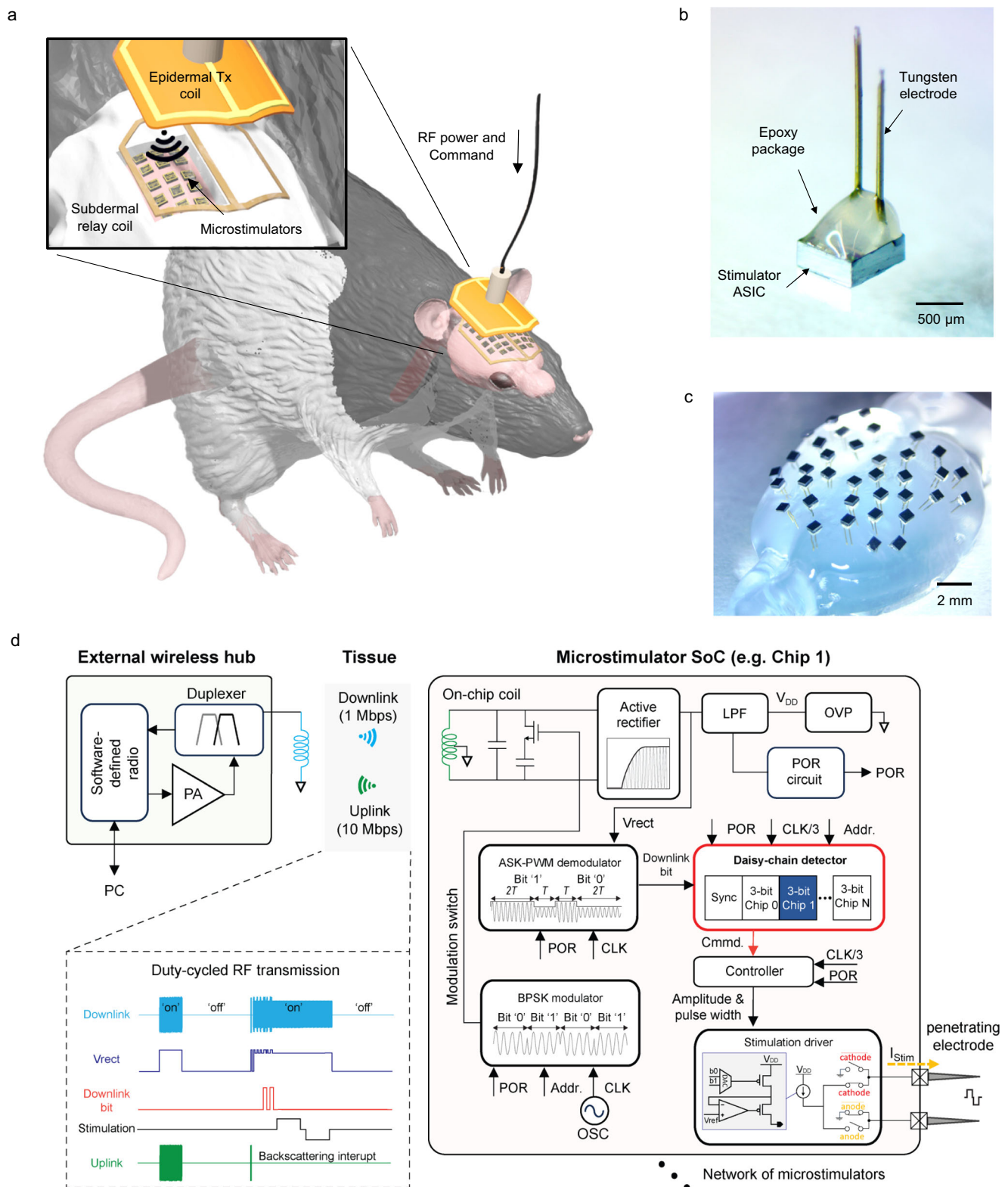
Overview of implantable wireless microstimulators—circuits and communication

The primary achievement of this work is the development of a wireless neuroengineering solution that leverages unique SoC microchip implant circuitry for remote RF programming, to rapidly program electrical stimulation parameters in a mere 3 microseconds per chip. This feature allows RF energy and instruction sets to be delivered as impulses and then switched off, energizing stimulators only during active current injection. The unique strategy reduces the average RF power by more than an order of magnitude compared to continuous RF radiation, an achievement that alone separates this work from previous studies that deploy methods of continuous energy delivery^{35,37,41}. As a proof-of-concept testbed for the advanced technology, we used the rat model, a stepping stone for future large animal experiments where the advantages of our distributed microstimulator interface can be fully leveraged. As such, this work does not purport to be a basic neuroscience study, but rather a demonstration of how the wireless operation of ensembles of spatially distributed microstimulator implants can affect behavior in freely moving rodents.

Figure 1a illustrates the principle of the wireless microstimulator implant for chronic experiments in a rat model. A population of spatially distributed autonomous microchips is deposited on the cortex of the animal and energized from epidermal and subdermal RF coils by near-field inductive coupling. Two coils are specifically designed to match the rat’s cortical geometry, covering an area of 8 mm $\times$ 13 mm on the cortex. Simultaneously, the downlink also transmits stimulus-specific instructions to the network. For delivery of intracortical currents, pairs of tungsten microwires separated by 100 μm were affixed to the epicortical chips in a post-process step (Fig. 1b, Supplementary Fig. 1). An example of an ensemble of spatially distributed functional stimulator chips embedded in an agarose brain phantom is shown in the photo of Fig. 1c to illustrate the autonomy of their individual placement.

Figure 1d shows the transcutaneous near 1 GHz RF link (downlink and uplink), a schematic of the timing diagram, and a block circuit description of an individual microchip. The rationale behind choosing the frequency near 1 GHz is described in Supplementary Note 1. The downlink signal from the external wireless hub simultaneously delivers RF energy and data by employing amplitude shift keying pulse width modulation (ASK-PWM). In ASK-PWM, the relative durations of high and low amplitude RF signals encode bits 1 and 0 at a data rate of 1 Mbps, as illustrated in the dashed box in Fig. 1d and Supplementary Note 2. This approach makes on-chip downlink bit recovery tolerant to variations in the free-running oscillator frequency^{37,40}. The ASK-PWM demodulator on the chip decodes the envelope of the rectified incoming RF signal (V_{rect}) to extract downlink bits, which is further analyzed in the daisy-chain detector.

In the daisy-chain, register-mapped communication scheme, the downlink bits consist of a ‘sync’ sequence followed by serialized commands for all target chips, without requiring any chip address information or enable sequence thereby offering major advantages



over e.g., our previous call-and-response approach³⁷ (Supplementary Note 3). The daisy-chain detector (shown in the red rectangle within the Microstimulator box of Fig. 1d) monitors the downlink bits generated by the demodulator. Upon detecting the correct 'sync' sequence, the detector identifies the N -th command (three bits per chip) assigned to a specific chip with the unique address N (detailed in Supplementary Note 4). The address, unique to each chip, is predefined by laser-ablating metal fuses in a chip postprocessing step³⁹. In

this method, a three-bit downlink instruction delivered in mere 3 μs programs a specific stimulation waveform for an individual chip; hence, in principle, arbitrary space-time patterns of microstimulation can be generated by the scheme for up to a thousand chips within 3 ms.

The digital controller uses the three-bit command from the daisy-chain detector to configure the stimulation waveform generated by the current source driver, adjusting both amplitude and phase duration. The current amplitude is controlled by adjusting the DAC output

Fig. 1 | Distributed microstimulator system. **a** Schematic showing a population of microchips implanted on rat cortex together with epi- and subdermal coils for resonant near-field RF energy transfer and delivery of stimulation commands. **b** Microphotograph showing a stimulator ASIC with integrated tungsten electrodes to enable intracortical microstimulation. **c** Photo of microchips inserted into a brain phantom made of agar to test the distributed wireless interface on benchtop. **d** Design of microchip circuitry and RF daisy-chain communication scheme for low-duty cycle operation. The downlink delivers RF power and commands at 1 Mbps to specific microchips to select one of the preprogrammed current waveforms. The ASIC consists of power harvesting, data recovery, daisy-chain detector, controller,

and stimulation circuit blocks. Power-on-reset triggers a BPSK modulation circuit to report its unique address, driven by a free-running oscillator. The inset shows the waveform of the downlink and uplink, rectifier output voltage (V_{rect}), demodulated downlink bit, and stimulation. The plot illustrates how the downlink is 'on' for a short period of time to deliver RF energy to the chip – only when needed resulting in duty-cycled RF transmission hence lowering average RF exposure. Abbreviations: V_{DD} : voltage supply, OVP: over voltage protection, POR: power-on reset, LPF: low pass filter, OSC: oscillator, CLK: clock, Addr.: address, Cmmnd.: command, PA: power amplifier, PC: personal computer, BPSK: binary phase shift keying, V_{rect} : rectified downlink signal.

voltage through b0 and b1 (Fig. 1d), while the phase duration is set by toggling switches for the cathode and anode. As the digital controller generates a single-shot stimulus, the downlink frequency determines the microstimulation frequency. The programmable operational amplifier current source driver delivers a predefined current across the tissue, alternating between the cathode and anode to ensure active charge balancing and prevent net charge injection over time (Supplementary Fig. 2). A feedback loop with a reference voltage (V_{ref}) ensures consistent current delivery to tissues, even when the impedance of the tungsten electrodes varies.

The integrated circuit of a current-injecting microchip also contains a binary phase shift keying (BPSK) modulator for backscattering (uplink), whose purpose is to report the unique address of each chip to the transmitter hub upon chip activation (turn-on). Note that, as the clock frequency of a chip can be extracted from the received BPSK uplink data, the data can be used as an indicator for the power status onboard a chip. Further details on the on-chip circuits can be found in Supplementary Note 5. The supporting external RF transmitter electronics in our experiments included a software-defined radio (SDR) to generate the downlink commands and to collect the uplink signals, an RF power amplifier, and miscellaneous microwave components.

Programming and Near-field RF Powering Microstimulator Ensembles

To ensure safe RF power delivery across ensembles of spatially distributed microchips, we considered the likely variations in the magnetic field across the total implant area and the constraints on energy harvesting. We implemented the stimulator circuit of Fig. 1d in silicon dies with sizes of 300, 400, and 500 μm , fabricated in the 65 nm low-power RF CMOS process. We measured how the size of the microcoil on the die affects the incident transmitting (Tx) power requirements (Fig. 2a). To maximize the available on-chip charge, we used an unregulated voltage supply for the overall circuit where an overvoltage protection (OVP) diode sets the upper limit of V_{DD} . Thus, the RF power affects the harvested power level as well as the voltage supply so that an increase in RF power causes an increase in the clock frequency of the free-running on-chip oscillator. This test also suggests that even smaller microstimulators can operate in a similar fashion albeit with increased RF power. In the animal experiments, we primarily used the slightly larger 500 μm size chips due to the availability of larger numbers of chips and higher fabrication yield in microwire integration.

Three-bit commands from the downlink (seven types, 6 types for single shot stimulation, 1 type for continuous 100 Hz stimulation, Supplementary Table 2) determine the current amplitude and pulse period of injected current as shown in Fig. 2b, where each command corresponds to the delivery of a specified amount of total charge (Fig. 2b inset). Due to the unregulated voltage supply, the current amplitude increases with respect to the incoming RF power as shown in Fig. 2c. However, since the clock frequency also increases with RF power, the stimulation period decreases. As a result, the two effects nearly cancel out so that the increase in injected charge (e.g. for Command 3) is up by only 13.7% despite a tenfold increase in incident Tx power (the inset of Fig. 2c). As an example, Fig. 2d illustrates the total injected charge generated by Commands 6 (maximum) and 1

(minimum), with values given in Supplementary Table 3. Note that, in this case, if the Tx power were to increase tenfold, the increase in delivered maximum charge was measured as only 38.8%, demonstrating the moderation of RF power level variations in the circuit design notwithstanding the choice of the simple unregulated voltage source.

The free-running oscillator in the microstimulator offers the benefit of low power consumption and a small footprint. Still, wafer-level variations in semiconductor fabrication can lead to clock frequency variance across an ensemble of RF-powered devices^{37,40}. Our ASK-PWM downlink protocol (Supplementary Note 2) is tolerant to anticipated clock variance; a feature validated by varying the downlink data rate as shown in Fig. 2e,f. The results demonstrate that the ASK-PWM demodulator can achieve a 100% success rate provided that the downlink clock variance is less than 23%, a value well within the range of observed clock frequency variance in our chips. The downlink protocol also allows the generation of a wide range of stimulation frequencies, here up to 2 kHz even if such frequencies might be well in excess of typical pulse repetition rates for neuromodulation. Even for such high-frequency stimulation, however, the amplitude of the injected pulsed current remains rather constant as shown in Fig. 2g,h where only the incident Tx power determines the amplitude of the current (Supplementary Table 4).

Since the period of active current injection in most neuromodulation schemes is relatively short, on the order of 100 μs ^{26,27}, the wireless RF energy transfer can likewise be kept short to reduce the average RF power low. Figure 2i demonstrates such low-duty cycle RF transmission by displaying the 100 Hz biphasic stimulation waveform with 100 μs and 500 μs per phase generated by a downlink at 4.6% and 14% duty cycle, respectively. Crucially, such a short-period downlink command can control multiple devices simultaneously and reliably, as demonstrated by benchtop measurements in Fig. 2j with five chips. In this example, six different downlink commands are transmitted sequentially, each command generating preprogrammed values of current amplitudes and periods per phase across the set (downlink sequences are depicted in Supplementary Fig. 3).

Ensembles of intracortical microstimulation on Benchtop and in acute in vivo rat model

After attaching a pair of tungsten microwires (Fig. 1b) onto each ASIC die for focal intracortical current injection (see Supplementary Fig. 1 for microwire post-process integration procedure), we first measured the pulsed current injection on the benchtop both by individual as well as a population of 24 microstimulators. Using phosphate-buffered saline (PBS) as a substitute for conducting tissue as illustrated in Fig. 3a, we first measured the RF-powered injected current using a sensing resistor in series as shown in Supplementary Fig. 4. The biphasic currents and applied voltages for commands 1, 2, and 3, generated by the ASIC, are shown in the top row traces of Fig. 3b (averaged over 60 measurements), where the pulse shapes include finite effects from the capacitive interface between the tungsten electrodes and the saline electrolyte. Data for commands 4, 5, and 6 are provided in Supplementary Fig. 5. Based on each specific wireless command, varying amounts of charge are delivered by the chips as

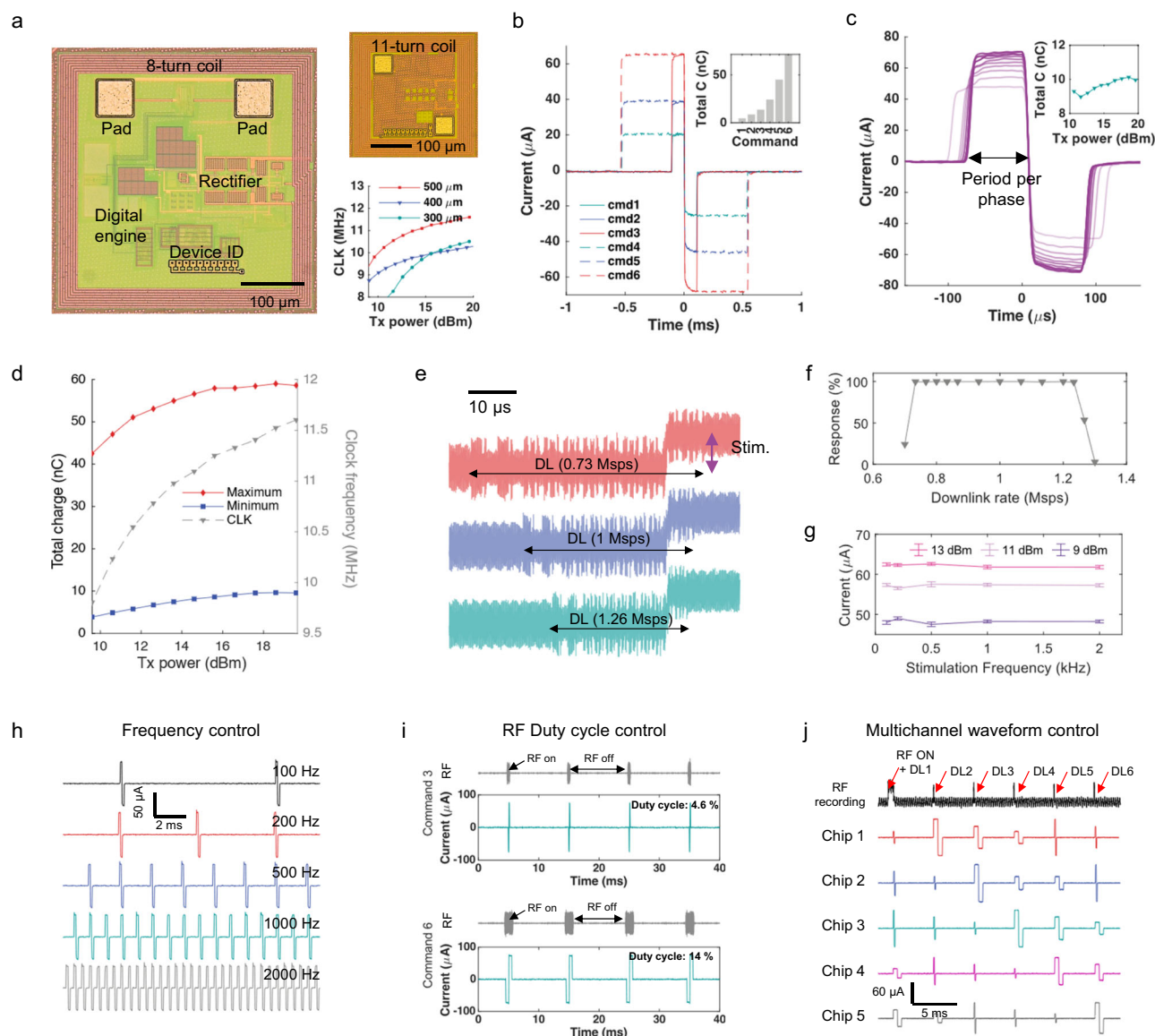


Fig. 2 | Properties of microstimulator ASICs. **a** Microphotograph of two stimulator chips of $400\ \mu\text{m} \times 400\ \mu\text{m}$ and $300\ \mu\text{m} \times 300\ \mu\text{m}$ size, respectively. The inset shows how on-chip clock frequency depends on the incident Tx power (benchtop test). **b** Current waveforms generated by six distinct downlink commands to vary both the amplitude and pulse width. The inset shows the total injected charge per specific command. **c** Expanded time axis of the stimulation waveform for a $200\ \mu\text{s}$ pulse width ($100\ \mu\text{s}$ per phase) at varying Tx power levels. The inset shows the total injected charge as a function of the Tx power. **d** Dependence on Tx power level for a maximum (Command 6) and a minimum (Command 1) of charge injection and on-chip clock frequency, respectively. **e** Downlink (DL) waveforms transmitted at different data rates ($0.73\text{--}1.26\ \text{Mpsps}$) and the stimulation event generated as a response to the commands for chip address 6. **f** Tolerance to a wide range of downlink rates when measured as a response (stimulation) to downlink commands

across 1000 trials. **g** The amplitude of current injection as a function of Tx RF power and a wide range of stimulation frequencies, demonstrating stable current injection up to $2\ \text{kHz}$ ($100\ \mu\text{s}$ per phase). Data points present average values, and error whiskers show average $\pm$ standard deviation ($n = 100$). **h** Control of the stimulation frequency directly by downlink transmission frequency. **i** Waveforms of the RF downlink and injected stimulation, respectively, demonstrating low-duty cycle RF transmission at $100\ \text{Hz}$ frequency. The duty cycle is lower for stimulation of $100\ \mu\text{s}$ per phase (top, Command 3) case, compared to $500\ \mu\text{s}$ per phase (bottom, Command 6). **j** Programmability of temporal stimulation across a five-chip population; a single downlink sequence determines the stimulation amplitude and period per phase simultaneously for all chips. In these experiments, the stimulation was delivered into $14\ \text{Kohm}$ load resistance. Abbreviations: ID: identification, CLK: clock, cmd: command, C: charge, stim.: stimulation.

shown in the bottom row data of Fig. 3b. While a finite charge imbalance remains, this amount is small relative to the total injected charge (generally less than 2%) because of active charge balancing. We then used a custom (wired) 24-channel planar recording electrode (inset of Fig. 3a), to illustrate the generation of arbitrary spatiotemporal patterns by a 24-element wireless network of microstimulators, here dynamically writing out the sequence of a series of letters spelling BROWN in Fig. 3c.

We next implanted six chips into the cortex of a rat to assess the in vivo wireless stimulation capability while the animal was under

anesthesia. Given the large parameter space in terms of all possible combinations of stimulation patterns of variable current amplitude, pulse width, and frequency, we did not aim in this paper at a detailed neuroscience study. Rather, we aimed to test the functionality of the multipoint wireless microchip system by selective sampling of the multidimensional stimulus parameter space. A wired Utah-type multielectrode array (penetrating MEA, channels 1–16) was implanted adjacent to the wireless stimulating microchips (Fig. 3d) to monitor both the current waveforms in cortical tissue and to record the waveforms of evoked neural responses. Given that the distance from

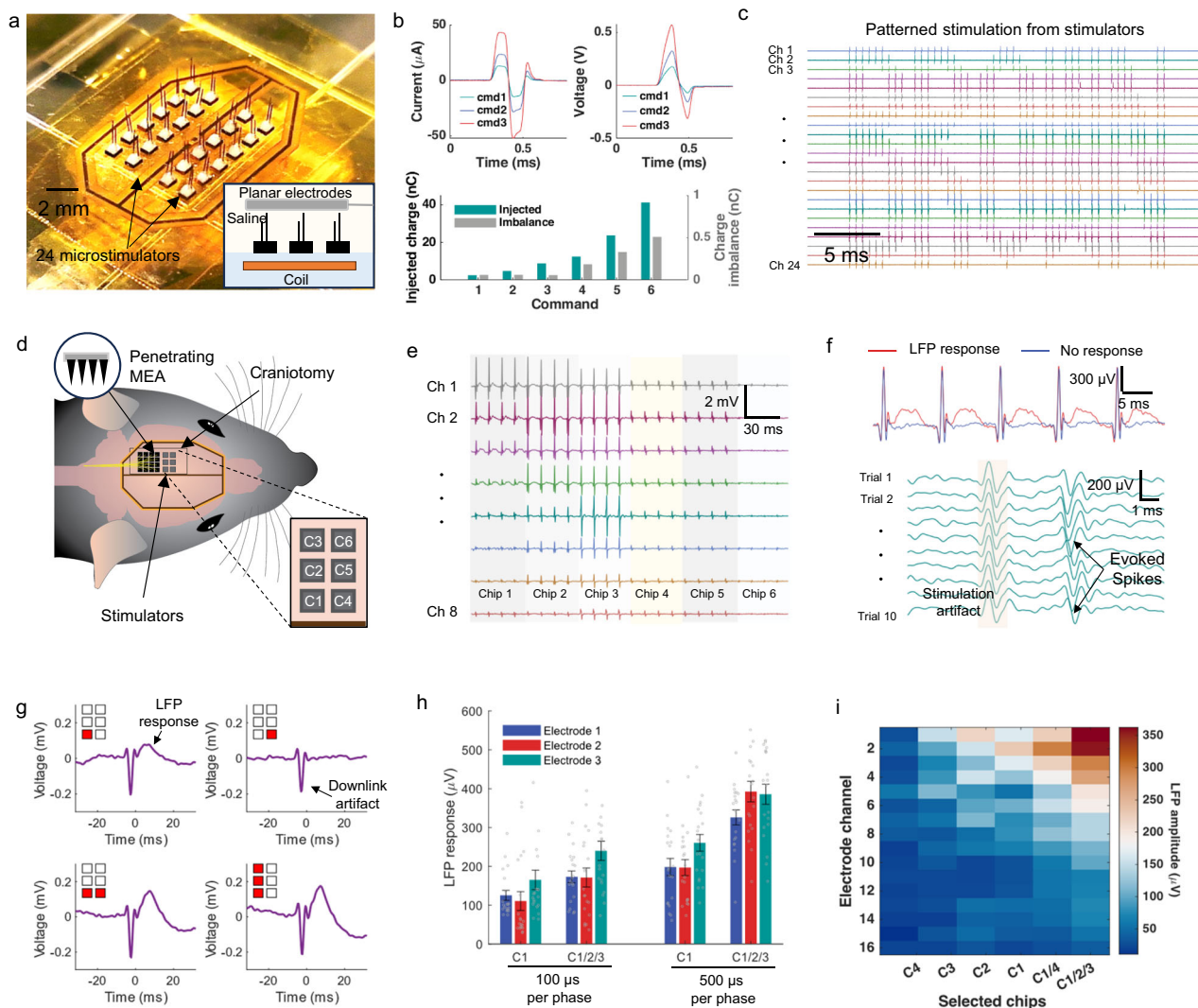


Fig. 3 | Demonstration of pattern stimulation by the wireless stimulator network on benchtop and in-vivo acute experiments. **a** Immersion of microchips in transparent saline. The inset shows the location of the relay coil (bottom of the dish) and a 24-element planar recording array with its size matching the coil underneath. **b** Voltage and current waveforms of stimulation injected by the microstimulator into the saline through a pair of tungsten electrodes (top row). The bottom row shows injected total charge and charge imbalance depending on the stimulation waveforms (all plots averaged over 60 trials). **c** Spatiotemporal stimulation waveforms generated by 24 microchips to inject a current pattern spelling BROWN. **d** Microchips implanted into a rat cortex in acute experiments (six chips C1 to C6, and the 16 channels penetrating MEA, separation of adjacent silicon needles is 400 μm). **e** A plot generated by selecting each of the 6 stimulators and injecting current, captured by 8 recording channels in MEA as the ‘stimulation artifact’ (Command 7). **f** LFP responses evoked by microstimulation (red, top) compared to the absence of response observed through the same electrode when

repeated stimulation led to apparent neuronal fatigue (blue, top). The plot at the bottom displays the evoked neuronal spike responses following each stimulation event. **g** Averaged LFP responses and downlink artifacts caused by rapid RF power switching during duty-cycled RF transmission, here for chips delivering currents with 100 μs period per phase (Command 3, 10 trials, bandpass-filtered from 60 to 2000 Hz). **h** Example showing the dependence of LFP response amplitudes on the stimulation period and the specific set of chips selected for stimulation. Bars present average values and error whiskers show average $\pm$ standard error ($n = 20$, dots for individual data points). **i** Composite heat map of LFP response amplitudes induced by selected chips (500 μs period per phase, Command 6) and measured across 16 recording electrodes to demonstrate both local and broad cortical modulation (averaged over 20 trials). Maximum Tx power of 20 dBm (100 mW) was applied in the acute animal experiments. Abbreviations: MEA: microelectrode array, LFP: local-field potential.

each current injecting microchip relative to a given MEA recording electrode varied, we could collect the matrix of data in Fig. 3e to verify that the wirelessly controlled address selection works reliably for an in vivo implant.

We probed the effect of focal microstimulation in selected locations of the cortex while actuating single or multiple microchips, achieving both local and broad cortical modulation. For example, when stimulating the cortex at 100 Hz using a given single microchip, we recorded both evoked local field potentials (LFPs) and extracellular spike activities (Fig. 3f). As shown in Fig. 3g, current simultaneously

injected by chips 1, 2, and 3 evoked higher amplitude LFP responses compared to that by single chip 1 alone. In addition, a longer period stimulation (500 μs current pulse vs. 100 μs pulse) evoked larger LFPs presumably because of the increased total injected charge as shown in Fig. 3h⁴². The results of the paired t-test analysis comparing conditions in Fig. 3h are summarized in Supplementary Table 5, which shows that both the multichip stimulation and the period of stimulation have statistically significant effects. These results also suggest that an equivalent level of cortical neuromodulation can be evoked by short-period stimulation delivered by multiple chips when compared to a

longer stimulus by a single chip. Note that the former case corresponds to a lower average applied RF power due to the lower duty cycle. Furthermore, temporally simultaneous stimulation from multiple chips generated robust LFP responses across a wider cortical area as illustrated in Fig. 3i and Supplementary Fig. 6. Here, stimulating the cortex by chips 1, 2, and 3, the recordings from the MEA displayed distributed LFP responses larger than 100 μ V at up to ten electrode sites whereas only five MEA electrodes could record comparable LFP responses from stimulation by chip number 1 alone. In these experiments, the downlink artifact, generated by rapid RF power switching during duty-cycled RF transmission, measured approximately 200 μ V with a full width at half maximum (FWHM) of around 2 milliseconds. This artifact saturated the amplifier but was later bandpass filtered, resulting in the artifact waveforms shown in Fig. 3g.

Multipoint intracortical microstimulation in chronic in-vivo rat model

For a demonstration of spatially distributed intracortical electrical microstimulation in an awake, freely moving animal, we implanted 30 microchips on the surface of the rat cortex together with a resonant RF energy-transferring subcutaneous relay coil (Fig. 1a). The chips were distributed across the surface of motor and sensory areas, each chip equipped with a cortex-penetrating microwire electrode pair. The location and distribution of the chip population were later imaged by postmortem micro-computed tomography (CT) scans acquired more than three months after surgery (Fig. 4a and Supplementary Fig. 7). The external Tx coil was mounted onto the head of the animal using silicone adhesive enabling the rat to move freely within a 33 cm $\times$ 33 cm $\times$ 42 cm sized enclosure which was instrumented for a behavioral lever-pressing task (Fig. 4b and Supplementary Fig. 8). The lightweight Tx coil (170 mg) was connected via a thin microcoaxial cable to an RF duplexer, an amplifier, and the SDR unit for RF power delivery and command/communication. Details of the hardware and the behavioral test setup can be found in Supplementary Figs. 9, 10 and Methods.

We first targeted the right hemisphere of the motor cortex and delivered current stimulus while recording any overt animal responses with a video camera. The current injection of 120 μ A (peak-to-peak) by a single microdevice was sufficient to evoke clear head movement as shown in Fig. 4c and Supplementary Video 1. We used high-frequency burst stimulation at 400 Hz (500 μ s per phase, 0.25-s pulse train duration), fast on the scale of a neuron's response time but a rate, which has been shown to evoke robust muscle twitch in rodents in other intracortical work^{43,44}. 'pulse train duration' refers to the time period during which the 400 Hz stimulation was delivered, resulting in a total of 100 current stimulation events in this case. The inset in Fig. 4c shows the location of the particular chip in the motor cortex and its position relative to the perimeter of the relay coil. Figure 4d plots the measured head angle changes during stimulation events, showing the stimulation evoked a tilt in the head angle and the subsequent corrective/reactive action by the animal then decreased the head angle over several seconds.

We then selected a set of three chips distributed across both the motor and sensory cortex (Fig. 4e). High-frequency stimulation (400 Hz, 500 ms pulse train duration, and 500 μ s per phase) evoked readily measurable high-frequency oscillatory motion of whiskers, as seen in Supplementary Video 2; the evoked response was also measured as post-stimulus changes in the whisker angle (Fig. 4f). An expanded view of high-frequency oscillations of whisker movement is shown in Fig. 4g. The result suggests that spatial summation of local currents injected from the three chips stimulated the animal's sensorimotor whisker circuit and triggered a corresponding motor response. To allow sufficient time for the cortical network to reset between stimulus events and to reduce the average RF power in each trial, we reduced the effective duty cycle to 4.37% for both head and

whisker movements by incorporating resting periods of 4 s and 8 s, respectively.

Further exploring the neuromodulation capabilities of the microchip implant, we designed a behavioral experiment where we trained the animal in a two-lever pressing task. The aim was for the animal to report percepts evoked by current stimuli (Fig. 4h). If the rat detects a stimulus, it is expected to press the left lever; if it does not detect the stimulus, it has been trained to press the right lever in both cases to receive a reward (Supplementary Video 3). A sound cue was played to initiate the report. When stimulating all thirty chips simultaneously at 50 Hz frequency, the animal successfully performed the detection task to achieve a 96% rate for the correct lever pressing response. When only two chips from the set were chosen—stimulating the sensory cortex (N2) or motor cortex (N'2) (see positions in Supplementary Fig. 11)—the correct response rates were reduced to 86% and 62%, respectively (Fig. 4i). The difference suggests that the animal's experiential sensation, that is its perception of the stimulus, depended on the cortical location where the focal current was injected. The kinematic response times, i.e. animal's reaction times for hits and correct rejections, are shown in Fig. 4j. There was a statistically significant difference in the response times for hits and correct rejections when the 30 microstimulator was actuated, perhaps indicating that the time between the animal's perception and subsequent reporting depended on the strength of the sensation induced by the current stimulus.

Finally, we tested whether the animal could perceive stimulation delivered at a low-duty cycle RF transmission—a key feature of our system, even when delivered by a single microstimulator within the set. In this experiment, we measured the correct response rate in the detection task when stimuli were delivered at two frequencies, 50 Hz and 20 Hz, and two pulse periods per phase, 100 μ s and 500 μ s (Commands 3 and 6). As shown in Fig. 4k, with a stimulation frequency of 50 Hz and a phase period of 500 μ s, the corresponding RF duty cycle was only 9.5%, yet the animal still achieved a 90% correct response rate. Even when the stimulation frequency was reduced to 20 Hz and the period to 100 μ s per phase, resulting in an RF duty cycle of only 1.4%, the animal managed an 82% correct response rate. This suggests that even modest charge injection under low RF duty-cycled transmission can evoke perceptual experiences detectable behaviorally. These in-vivo experimental results demonstrate the versatility of our wireless system, which opens the doors to seeking for optimal stimulation strategy for neuromodulation through parameters such as patterned channel selection, current stimulation period, amplitude, and frequency while minimizing RF energy exposure.

Stability of the wireless link in chronic implants in freely moving rat

The RF powering of and backscattered signals from the chronic wireless implant involved inductive coupling via the wireless 3-coil system of Fig. 1a, namely the Tx coil above the animal's skin, the subcutaneous relay coil above the skull, and the chip microcoils located on the cortical surface. Using a micro-CT scan to model the tissue layers for electromagnetic stimulation, we assessed the efficiency of wireless power transfer based on the location and angle of the microstimulator, as shown in Supplementary Figs. 12, 13. When the on-chip coil is parallel to the relay coil, the estimated efficiency between the Tx coil and the on-chip microcoil ranges from -19.22 dB to -20.91 dB. However, due to limited precision in fabrication and implantation methods, the angle of the microstimulator may vary, affecting wireless efficiency. In evaluating the implantation angles of 30 microstimulators, we found that angular rotation of the stimulator could cause up to an additional 2 dB loss. Despite these variances, and other factors such as impedance mismatching, limited coil quality factors, and semiconductor process variations, transmitting 18 dBm of RF energy was sufficient to activate all chips

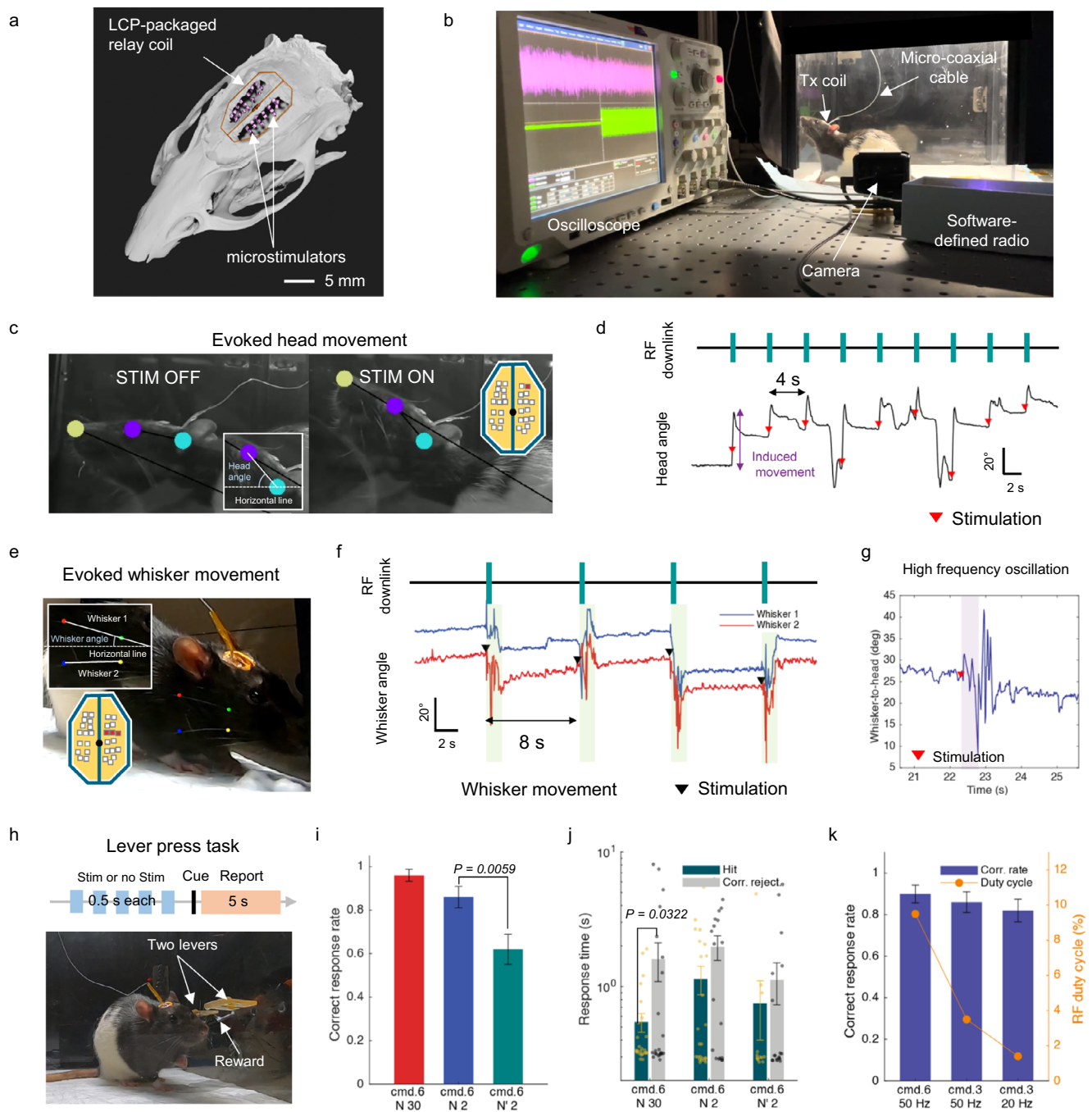


Fig. 4 | Microchip implants in freely moving chronic rat model. **a** Postmortem micro-CT scan image of the animal's head showing the location of the relay coil and microstimulators (additional views in Supplementary Fig. 7). **b** The experimental setup includes a custom enclosure allowing the rat to move freely, with the Tx coil attached to the head using silicone adhesive; supporting instrumentation includes a SDR, an oscilloscope to monitor the downlink, and a camera. **c** Captured video images showing the head position before and following stimulation. The inset shows the location of the particular microchip (in red) from a population of 30 chips delivering stimulation (400 Hz, 250 ms stimulation pulse train duration, and 500 μ s per phase, the black dot in the center is aligned with the bregma). **d** Timing and duration of the RF downlink to trigger focal microstimulation of the motor cortex and measured head angles relative to the horizon after each stimulation event. **e** Photo of animal and the two target whiskers: whisker 1 labeled with red and green dots, and whisker 2 labeled with blue and yellow dots. The inset shows three microchips (each injecting pulsed current at 400 Hz, 500 μ s per phase, and 500 ms train duration). **f** Timing of the RF commands and stimulus-evoked whisker movement; whisker angles are measured with respect to the horizon. **g** Post-

stimulus, high-frequency oscillatory motion of evoked whisker response. The purple-shaded area on the graph represents the duration of the pulse train. **h** The timing diagram for the choice lever-pressing task and the lever setup. If the animal perceives the current stimulus, it should report the percept by pressing the left lever; otherwise, it should press the right lever. An audio cue marked the start of a report session. **i** Correct response rate based on the number of microstimulators delivering current stimulus (the location of N2, N'2, and N1 is shown in Supplementary Fig. 11, the trial number is 50, stimulus parameters: 50 Hz, 500 ms pulse train duration, 5 train repeats). **j** Response (reaction) time for 'Hit' and 'Correct rejection' in relation to audio cue, respectively. **k** Dependence of correct response rate and RF duty cycle on the stimulation frequency and the period of each stimulation phase (single chip stimulation, N1, 500 ms train duration, 5 train repeats). In **i**, **j** and **k**, Bars present average values and error whiskers show average $\pm$ standard error ($n = 50$). The p -values shown in the plots were obtained from paired t-tests indicating significant differences ($p < 0.05$). The Tx power was 20.15 dBm in all chronic experiments. Abbreviations: LCP: liquid crystal polymer, Corr. reject.: correct rejection, Corr. correct.

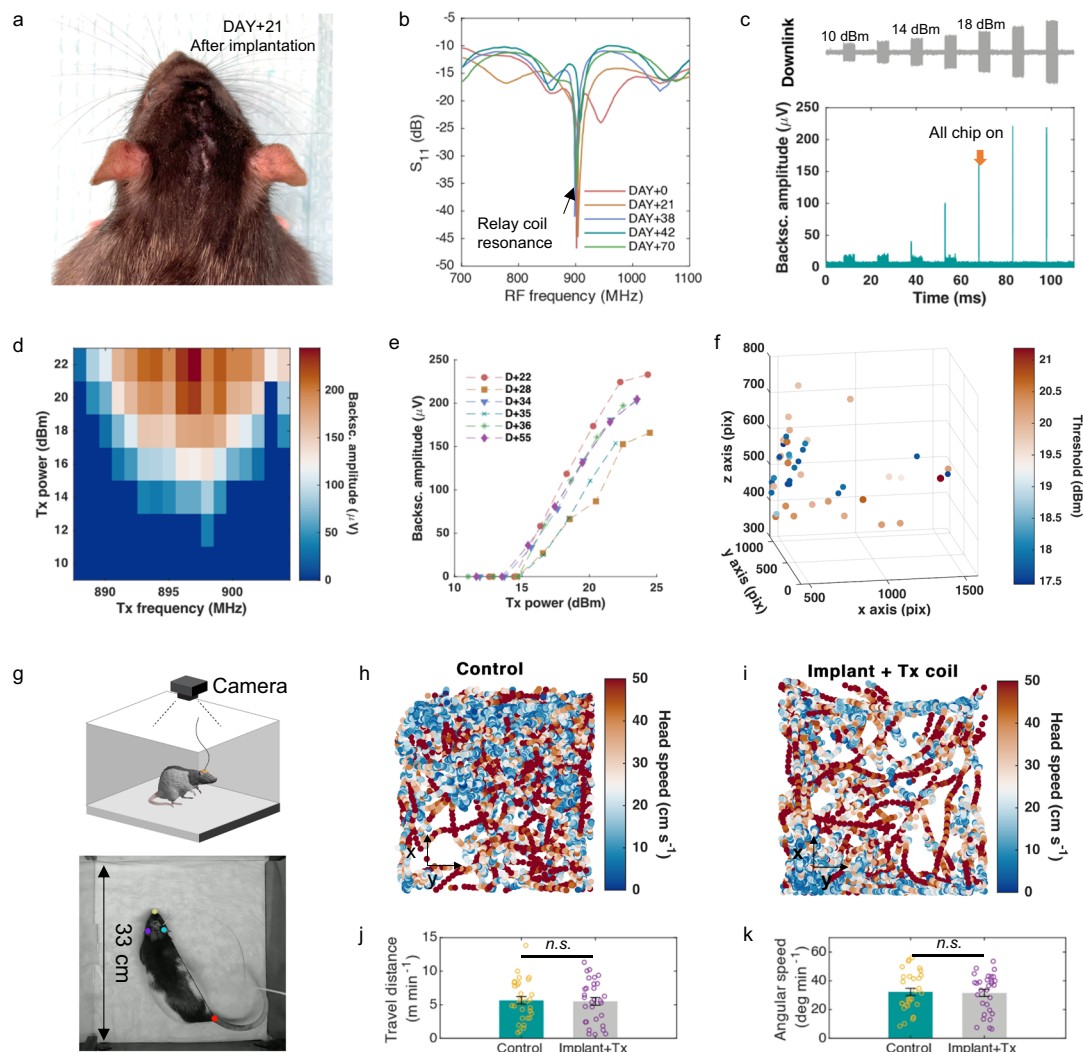


Fig. 5 | Stability of the 3-coil wireless link in the chronic rat implant. **a** A photograph of the animal's head 21 days after the implantation of the relay coil and 30 microchips. **b** Measured S-parameter (S_{11}) of the Tx coil attached to the head of the target animal, showing the resonance of the implanted relay coil over two months. **c** RF downlink signal with varying power amplitudes (top) and amplitudes of backscattered signals from microchip population displaying a threshold Tx power of 18 dBm (bottom). **d** Backscattered signal amplitudes as a function of Tx power level and Tx frequency. **e** The amplitude of backscattered signals recorded up to 55 days following implantation and its dependence on Tx power. The measurement was obtained from the single animal. **f** Selected x-y locations of the animal freely moving in the enclosure with corresponding threshold Tx power (additional coordinate plane views in Supplementary Fig. 15 and video capture examples in

Supplementary Fig. 16). **g** Schematic and photo captured by camera used for motion tracking inside the 33 cm $\times$ 33 cm $\times$ 42 cm sized enclosure. **h** and **i** The location and speed of the animal's head tracked for 30 min for both a control rat and the rat with implants and the Tx coil attached to its head, respectively. **j** and **k** Distance traveled per minute and head angular speed per minute, both averaged over 1 min, showing no statistically significant difference between the control rat and the rat with implants. Bars present average values and error whiskers show average $\pm$ standard error ($n = 30$, p value 0.85 and 0.83 for distance traveled and angular speed, respectively). The p -values were obtained from paired t-tests indicating no significant differences ($p > 0.05$). Circles represent each data point. Abbreviations: Backsc.: backscattering, n.s.: not significant.

inside the rat, as shown in Fig. 5c, where each chip needed to harvest at least -3 dBm for operation. We note that the optimal harvested power range for the microstimulator is from -3 dB to 0 dB for efficient performance, but the circuit can also operate with much higher received energy due to the OVP diode on the chip.

Neither the subcutaneous relay coil nor the chips were visible once the scalp was fully healed after 21 days postsurgery as shown in the photo of Fig. 5a. For hermeticity, the subcutaneous relay coil was encapsulated by a 225 μ m thick layer of LCP. Its resonant coupling characteristics could be measured by monitoring the relevant S-parameter of the external Tx coil, reflectance S_{11} , and a network analyzer. The value of S_{11} as a function of time is plotted in Fig. 5b to illustrate how the relay coil maintained a sharp resonance peak even 70 days after implantation (measurement setup in Supplementary Fig. 14, single subject). Before each S-parameter measurement, we

shaved the hair of the target area to ensure that hair regrowth did not interfere with the coil attachment or RF power harvesting stability.

We also examined the strength of the inductive coupling across the microchip population by varying the Tx power and measuring the amplitude of backscattered signals. When all chips were activated, the backscattered signals summed up to a short “reporting” pulse in Fig. 5c. Here, the total amplitude of the backscattered signal is proportional to the number of active devices and the Tx power, but is also affected by variations in the efficiency of the wireless link. We evaluated the effective efficiency of the 3-coil resonant system by varying the Tx frequency and measuring the backscattering amplitude, where Fig. 5d shows the resonance at 898 MHz. We monitored the amplitude of the backscattered signal over two months and observed good stability of energy harvesting over time in the freely moving animal, as shown in Fig. 5e. Along with Fig. 5b, these results also indicate that LCP

packaging is impermeable to water or electrolyte intrusion at least over two months where such intrusion can lower the quality factor of the coil and/or cause a shift in the resonance frequency.

To test the possible variation in threshold RF power required to turn on all chips while the animal moved inside the 33 cm × 33 cm × 42 cm sized enclosure (Fig. 5g), we monitored backscattering signals at multiple x-y coordinate points. The results in Fig. 5f and Supplementary Fig. 15 show no measurable dependency of the threshold on the animal's x-y position. Rather, a more important factor was the variation in its behavior which includes scratching, standing, exploring, and nesting (video capture shots in Supplementary Fig. 16). When the animal was still, the average threshold Tx power was 17.78 dBm. Those behaviors that involved direct contact of paws with the head and Tx coil caused the threshold to increase by up to 3.4 dBm. In this worst-case scenario, the misalignment of the coils led to a 2.1-fold increase in the required RF power. Even so, the average power values still remained well below the specific absorption rate (SAR) safety limit.

Video monitoring (Fig. 5g) also allowed us to answer two additional questions: 1) whether the multichip implant and the subdermal relay coil themselves might impair the motor activity of the animal, and 2) whether the microcoaxial cable from the Tx coil to RF electronics hindered its freedom of movement. Figure 5h,i show that the animal was able to move across the enclosure comparably to another animal without the implant (referred to as 'Control'). Furthermore, the differences in distance traveled per minute or head angular speed per minute between the two sets of data acquired from each animal were not statistically significant, as shown in Fig. 5j,k.

Discussion

In this paper, we have demonstrated the delivery of patterned electrical brain stimulation by a spatially distributed wireless network of implanted silicon microchips. The network can generate arbitrary space-time patterns of stimuli across multiple cortical areas as a chronic rodent implant. One key feature in the design of the wireless system is the RF command-and-control communication protocol which enables virtually instantaneous control of a population of microstimulators to generate complex patterns of current injection. Such spatiotemporal programming is possible under low-duty cycle RF conditions so that the average RF exposure to tissue remains low in a long-term implant. For a proof-of-concept demonstration, we implanted microchips in both anesthetized and freely moving rats to demonstrate evoked neural and behavioral responses to multipoint cortical neuromodulation. The wireless system was able to trigger head and whisker movements by a direct current stimulus into the motor cortex and reliably generate percepts by activating the underlying circuits in the sensory cortex while the animal was engaged in a trained behavioral task.

Since the local SAR regulatory limit is much higher than that for whole-body exposure^{45,46}, local power delivery in a duty-cycled transmission scheme can keep the RF exposure well below SAR limits by reducing the average transmitted power. In the chronic studies, we delivered up to 103.5 mW of peak Tx power directly into the animal's head. Supplementary Fig. 17 shows that the 3-coil system designed here for rodents has a peak spatial SAR value of 1 W/kg (averaged over 10 g of tissue at continuous 100 mW Tx power), an order of magnitude below the tissue safety threshold for human use of 10 W/kg^{37,45,46}. Yet, by lowering the operational duty cycle to 1.4%, the average RF power required to elicit a behavioral response was a mere 1.45 mW. This excitation was sufficient to achieve an 82% success rate in a lever-pressing task, even by a single microstimulator chip. Therefore, operating in low-duty cycle transmission mode yet providing effective neuromodulation, the corresponding SAR value could be reduced by another order of magnitude down to 0.0145 W/kg for a stimulation period per phase of 100 μ s and stimulation frequency 20 Hz (again 1.4% duty cycle).

In adapting the current system to larger animals, including non-human primates, the stimulation parameters—current amplitude, phase duration, and frequency—remain comparable to those used in rats, as evidenced by the large body of prior electrical neuromodulation studies^{4,5,13,19,21,47,48}. However, due to an increase in overall tissue thickness, the RF near-field coupling efficiency is reduced, requiring higher incident RF energy. Our electromagnetic simulation results, showing how the wireless efficiency varies with tissue thickness and the depth of microstimulators, are shown in Supplementary Fig. 18. In the case of thicker skin, fat, and skull (3 mm, 2 mm, and 4 mm, respectively), the wireless transfer efficiency can decrease by an order of magnitude, requiring up to 1 W of power transmission during active stimulation. In two previous studies in non-human primate and human subjects^{47,48}, applied stimulation duty cycles were approximately 9.3%, resulting in an expected average SAR of 0.93 W/kg. Therefore, even allowing for a tenfold decrease in the efficiency of near-field energy transfer in our system for thicker tissue in a primate (scalp, skull, dura)³⁷, reaching a SAR value well below even the low-tier threshold of 2 W/kg appears to be possible⁴⁶.

Some of the remaining challenges in adapting the advanced technology to large animal models and translation to clinical applications stem from the microfabrication and implantation of the wireless microchips. The miniaturized stimulators have a lateral dimension of 500 μ m, weigh approximately 300 μ g, and feature penetrating microwire electrodes with a length of 1500 μ m. The tungsten electrodes are not perfectly perpendicular to the microchip surface and the biomedical-grade epoxy applied for mechanical support forms a curved interface, factors that inhibit a close, flat contact between the stimulator and the cortex. To ensure safe implantation and long-term stable current delivery, these issues need to be addressed for translational use. Our plan is to improve electrode fabrication methods and utilize an implantation technique based on a bioresorbable polymer⁴⁹. Elsewhere, we have already demonstrated that atomic layer deposition (ALD) provides conformal, low-volume encapsulation for the longevity of the micro-devices, serving as an alternative to the parylene-C package used in this work⁵⁰. Furthermore, while tungsten electrodes have been utilized in studies for cortical interfaces for decades^{51–55}, the rigid, stiff tungsten electrodes can induce an inflammatory response due to the mechanical mismatch with the rat brain tissue⁵⁶. Flexible electrodes, using low-impedance materials, such as Poly(3,4-ethylenedioxythiophene) Polystyrene Sulfonate (PEDOT:PSS)⁵⁷ or carbon nanotubes (CNTs)⁵⁸, have the potential to provide better mechanical compatibility, deliver charge efficiently, and maximize the advantages of the free-floating form factor in our tetherless microstimulators⁵⁹. Lastly, we observed that our current source driver circuit occasionally showed an imbalance in the injected charge, reaching up to 3% of the total charge in the specific command, Command 5. For long-term use of our system, more sophisticated on-chip charge monitoring may help improve stability at the tissue-electrode interface.

In this work, we focused on demonstrating an innovative neurotechnology for multipoint space-time patterned electrical stimulation. While the rodent model is a useful testbed for assessing the viability of the spatially distributed autonomous wireless neuromodulation microchip system, it is limited in basic neuroscience value in this case. This is partly because the repertoire of behavioral responses in rodents is limited and partly because the cortical area is small. As a result, it is impractical to test the exceedingly large parameter set of combinations of chips, amplitude, pulse duration, and repetition rates that can be achieved through our distributed wireless technology. On the other hand, given a larger animal model with a well-known cortical mapping (e.g., retinotopy in the primate primary visual cortex²⁰), our system offers unique opportunities for the study and application of systematic patterned stimulation, as it can reach across large areas of the cortex.

In such models, possible optimal combinations of stimulation sites and parameters can be narrowed down based on the subject's behavioral response as feedback.

In summary, the wireless technology introduced in this paper presents multiple opportunities for basic neuroscience studies and possibilities for clinical neurotechnologies. The concept of spatially distributed autonomous ensembles of microstimulators eliminates the need for inter-node and power line connections prevalent in other current approaches, which are particularly prone to field-driven corrosion and can potentially damage tissue. In this regard, our approach may achieve greater stability compared to systems that rely on heterogeneous off-chip connections between an ASIC chip and a power source. In addition, the daisy-chain, duty-cycled energy transmission method can be applied to other excitation modalities, such as midfield energy transfer⁴¹, ultrasound energy transfer^{35,60}, magnetoelectric energy transfer³⁶, and optical energy transfer^{33,61}. Our system can be further developed as a fully wireless solution for small animal models even without the head-mounted Tx coil, by leveraging the technology described in⁶², which uses an RF frequency of 1.5 GHz, comparable to ours. In this scenario, although wireless efficiency may decrease by up to 7 dB—requiring 500 mW of RF energy—the tissue temperature increase is still expected to be less than 1 °C due to the low-duty cycle RF transmission. We note that most previous wireless systems rely on voltage supplies assisted by large storage capacitors with long charging times^{35,60}, which may require architectural adaptation for rapid on-off switching in the duty-cycled energy transmission.

The ability to deliver arbitrary spatiotemporal patterns of current stimulus over a wide parameter range with minimal latency can be particularly valuable in a closed-loop BCI system designed for the restoration of sensations^{63,64}. Likewise, the system could offer an alternative approach to the development of visual cortical prostheses with implanted microchips across the primary visual cortex generating phosphene patterns with input from external cameras⁶⁵. As such, our wireless microchip approach is quite general and potentially broadly applicable to biomedical engineering applications which can benefit from multipoint, patterned electrical stimulation of specific physiological targets in the body. These might include cardiac stimulators as multipoint pacemakers designed to correct specific circuit disorders and subdermal microchips to convey tactile feedback to amputees wearing prostheses, to name but two examples. In such applications, stimulation currents in the milliamperage range can provide versatile functions by exciting nerves from a distance. While modifying the current driver circuit could enable this capability in the future, energy harvesting might necessitate larger coil and chip sizes—even with the low-duty-cycle RF transmission that we have introduced through our work.

Methods

Microstimulator design and fabrication

We designed the microstimulator ASIC in TSMC's 65 nm mixed-signal/RF low-power CMOS process. The daisy-chain detector designed for low-duty-cycled RF transmission provides a core distinction from the previous design using on-chip address-based call-and-response communication presented in the previous work³⁷. After the chip had been fabricated, we integrated a pair of 50 µm diameter tungsten microelectrodes for intracortical access as described in Supplementary Fig. 1. We used silver epoxy (H20E, Epo-Tek) to first fabricate ball supports at the edge of electrodes with the curing temperature of 200 °C for 15 min. Subsequently, we placed the electrodes on the chip after applying additional silver epoxy to the 60 µm chip pads. Ensuring perpendicular alignment of the electrodes with the chip surface, we cured the epoxy at a low temperature of 80 °C for 8 h to prevent chip damage. For mechanical reinforcement and encapsulation, we applied biomedical-grade epoxy (M-31CL, Loctite) to the base of the electrode and the chip. Lastly, the tips of stimulator electrodes were inserted into

the polydimethylsiloxane (PDMS, Dow corning) polymer and then the other parts were coated with parylene-C to achieve better biocompatibility and further encapsulation. Chemical vapor deposition formed a 3 µm conformal coating of parylene-C on the microstimulators, except for the tip of microelectrodes.

Wireless coil fabrication

The Tx coil and the relay coil were designed using Ansys HFSS and fabricated with polyimide printed circuit boards and encapsulated with biomedical epoxy and LCP, respectively⁵⁰. After attaching resonance capacitors (0.8 pF in parallel and 1.2 pF in series) to the Tx coil using solder paste (SMD4300SNL10, Chip Quik), we applied a minimal amount of biomedical epoxy to encapsulate the Tx coil. For the relay coil, we used thermoplastic LCP packaging to ensure the long-term safety of the coil. This packaging used multilayer LCP films combining high (CTZ-50, Kuraray) and low (CTF-25, CTF-50, Kuraray) melting temperature variants to improve interlayer adhesion and minimize defects. The thermocompression process using a heating press (model 4386, Carver Inc.) initially involved pressing films at 250 °C with a pressure of ~250 kPa for 10 min to shape the top layer, then pressing at 284 °C with ~450 kPa for 30 min to seal the top and bottom layers after aligning the coil in between. After fabrication, the hermetically packaged coil was subjected to an accelerated aging test at a controlled temperature of 120 °C for 3 days to confirm that the coil exhibited no changes in resonance properties. Any infiltration of water lowers the resonance frequency and the quality factor of the coil, which was identified through S-parameter measurements using the network analyzer (E5071B, Agilent). After the implantation of the relay coil in the rat, we checked its resonance using a portable network analyzer (MS2036A, Anritsu) as shown in Supplementary Fig. 14.

RF hardware for downlink command generation and uplink monitoring

We performed experiments with the stimulator chips using an SDR (the Model Raptor by Rincon Research). For benchtop characterization, we used a single-turn printed circuit board including the Tx coil (9 mm × 9 mm) and wire-bonded chip pads to the board, connecting chips with a load resistor (14 kΩ) or electrodes inside the saline (Supplementary Fig. 4). The SDR, using both transceiver chips (AD9361, Analog Devices) and Zynq SoCs, generated an RF downlink at 898 MHz, which was further amplified by an RF power amplifier (ADL5605-EVALZ, Analog Devices). The downlink ASK-PWM waveform was programmed on MATLAB, transferred to SDR, and imported to the field programmable gate array of the SDR using the IIO Oscilloscope library, which is controlled by MATLAB with the SSH protocol. The downlink signal is delivered to the Tx coil through an RF surface acoustic wave duplexer (D5DA942M5K2S2, Taiyo Yuden), which also isolates the backscattered signals from the microstimulators. The SDR amplifies the backscattered signal, downconverts it from 928 MHz (backscattering band) to DC, and finally digitizes the signal at 30 MSa/s (12-bit). The digitized data was then transferred to a personal computer for address and clock data extraction through BPSK demodulation and for calculating backscattering signal amplitude (computation performed in MATLAB).

Animal surgery and device implantation

The microstimulators were sterilized using ethylene oxide before the implantation. We performed surgeries on two Long Evans male rats (wild-type, 500–700 g, the age of 16–30 weeks obtained from Charles River Laboratories) for acute evaluation and chronic experiments. After shaving the animal's hair, we secured it to a stereotaxic frame and sterilized the head skin. We then made a sagittal incision up to 5 cm long in the skin over the skull, followed by a craniotomy using a precision surgical dental drill. For the acute evaluation, we performed a craniotomy on the left hemisphere and implanted six microstimulators

and a 16-channel penetrating multielectrode array from Blackrock Neurotech. While delivering the stimulations through wireless microchips, the explorer Nomad system (Nomad, Ripple Neuromed, sampling rate of 30 KSa/s) captured the neural responses from the electrode array. Recorded signals were first filtered with a 60 Hz notch filter to remove environmental noise, and then filtered again with a bandpass filter with a low cutoff of 60 Hz and a high cutoff of 2000 Hz. This was done to capture both the post-stimulus evoked LFP responses and the stimulation/downlink artifacts. During the initial surgery, the animal was anesthetized with 2–3% isoflurane but once the implantation was over, we switched anesthesia to ketamine/xylazine mixture (80 mg/kg and 5 mg/kg, respectively) to monitor the neural response better.

In the case of chronic surgery, we performed a bilateral craniotomy and manually implanted fifteen microstimulators on each side of the hemisphere. For the implantation, we adjusted the location of the implant to avoid the blood vessels. After the implantation, the open craniotomy was covered with the LCP-packaged relay coil, and we applied surgical silicone adhesive first at the edge of the craniotomy and then covered the relay coil edge with dental cement. Lastly, sutures closed the surgical area, and the animals were allowed to recover from anesthesia before being transferred to the cage facility for post-surgical monitoring. Animals were continuously monitored throughout the studies and euthanized with pentobarbital sodium according to approved methods upon study conclusion. The implantation surgery for the chronic experiment was conducted under 1–3% isoflurane. All research protocols were approved and monitored by the Brown University Institutional Animal Care and Use Committee. Additionally, all research activities were conducted in compliance with relevant NIH guidelines and regulations.

Microscale CT imaging

Micro-CT imaging was performed on a postmortem skull preserved in 4% paraformaldehyde using a SkyScan 1276, Bruker. The scan setting included an effective image pixel size of 10 μm , 1 $\times$ 1 binning, with a total of 1800 projection views made in a full 360-degree scan, along with an exposure time of 1250 ms. A peak source voltage of 100 kV and a current of 200 μA were used. The X-ray beam was filtered with a 0.5 mm thick aluminum filter and a 0.03 mm copper filter. Reconstruction was done using a Hamming filter ($\alpha = 0.54$) processed by the NRecon software (v. 2.0.0.5). We collected micro-CT imaging to verify the positions of the microstimulators and evaluate the status of the relay coil throughout this period. For that, the specimen was decapitated after being fixed in 4% paraformaldehyde 105 days after the implantation. Postmortem pictures are shown in Supplementary Fig. 19.

Design of the behavioral task and the apparatus

We built a custom enclosure with a motorized swivel using a 5 V stepper motor (28BYJ-48) and the driver board (ULN2003) controlled by Arduino. Since thin SMA cable (36 AWG, Alpha wire) cannot generate sufficient force for any shaft rotation, we manually controlled with rotation of the SMA connection using buttons, which are connected to a rotatory joint SMA adaptor (CR4794, Centric RF). On one side of the enclosure, we installed two levers for the detection and report task. The lever is made with photosensitive resin using a 3D printer (Mars 3, Elegoo) and connected to a snap-action switch. Levers were connected to Arduino boards for detection. We used MATLAB to collect information from the Arduino board during the behavioral task as well as to generate a downlink for intracortical microstimulation. For the lever task, once stimulation was delivered, the animal waited until it heard the report cue (1 kHz, 1 s, generated by MATLAB), and it pressed the lever to report within 5 s. When the animal pressed the correct lever, the liquid reward, 10% diluted condensed milk (California farms sweetened condensed milk), was given while the total amount of the liquid was limited to 20 ml

per day. If the animal pressed the lever when it was not reporting period, the initiation of the trial was delayed for 3 s to prevent the animal from randomly pressing the buttons.

Animal tracking and movement analysis

When stimulating the motor cortex or tracking the movement of the animal in the enclosure, we used two Hero 7 Black (GoPro 1080p HD 1920 $\times$ 1080, 30 or 60 FPS) cameras mounted to the top and side of the enclosure. Two cameras share the audio input (random noise generated from MATLAB) to synchronize the frame as illustrated in Supplementary Fig. 10. This sound was also captured by the oscilloscope (Tektronix MSO 5204), which monitors RF downlink in parallel. Using the oscilloscope and camera recording, we could precisely find the stimulation and downlink timing in the video. Tracking of the animal head's or whisker's position for either top-view or side-view videos was performed using the DeepLabCut package (Version 2.3.8)⁶⁶. Body features, such as ear, nose, whisker base and tip, were chosen for the tracking. The training was conducted separately for each camera view using a GPU (NVIDIA GeForce GTX 1070) with 200,000 iterations. The tracking session results were exported to Excel format, which included the X and Y coordinates along with the confidence value (Likelihood) for each data point. The data points with a confidence value greater than 99% were used for head tracking and analysis for head velocity, distance traveled, and head angular velocity. Meanwhile, coordinates with a confidence value above 70% were selected to capture the fast oscillatory movements of the whiskers with manual error correction applied.

Statistical analysis

Results in this work are presented as mean values $\pm$ s.d. (standard deviation) or mean values $\pm$ s.e.m. (standard error of the mean), which are described in the corresponding figure legends. Collected data are analyzed by MATLAB R2022a with custom code.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data supporting the findings of this study are available within the article and its supplementary files. Any additional requests for information can be directed to, and will be fulfilled by, the corresponding authors. Source data are provided with this paper.

Code availability

The codes for downlink generation and stimulation control are available within the article's supplementary files. Any additional requests for information can be directed to, and will be fulfilled by, the corresponding authors.

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Author contributions

J.L. and A.N. conceived the project. A.N., A.-H.L., and J.L. designed the animal experiments and methodology. A.-H.L. and J.L. designed and characterized the wireless ASICs and performed all experiments. A.-H.L., J.L., and A.N. wrote the manuscript. A.-H.L., J.L., V. L., L. L., and A.N. contributed to the data analysis and provided feedback on the manuscript.

Competing interests

The authors declare no competing interests.

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